

Task Brief Structure for Stateless Workers

What Nova should put in a brief to a worker that starts empty, and what the worker should send back. A recommended brief template and return template, eight sourced answers, a pre-send validation ladder, and the open questions. Fact-checked against primary sources.

0. Verdict on the framing

The question is well posed, with two corrections.

First, "optimal structure and wording" overstates what the evidence can deliver. The measured effects cluster into three tiers. Large, replicated, and mostly model-independent: whether the brief is *complete* (unstated requirements are followed ~41% of the time), whether acceptance criteria are *mechanically checkable* (the only thing that stops self-declared success), how many *constraints* it carries (compliance collapses multiplicatively past ~5–10), and whether the worker is told *what to do when information is missing* (otherwise it invents, 80–99% of the time). Medium and model-dependent: instruction *placement* relative to data, example count, and reasoning-before-schema ordering. Small or noise for frontier models: the choice between Markdown, XML, and labelled text; persona lines; tone; emphasis markers. Spend design effort in that order.

Second, the brief is not the only lever, and the evidence says it is not even the strongest one for the failure you most care about. No phrasing of a completion criterion reliably stops a worker from claiming success; what stops it is a check the worker does not control. The return-format recommendation (§2) and the pre-send validation ladder (§4) carry as much weight as the brief template (§1).

One thing in the framing is simply right and worth keeping: the stateless worker is exactly Anthropic's "colleague with minimal context" test made literal — "Show your prompt to a colleague with minimal context on the task and ask them to follow it. If they'd be confused, Claude will be too" ([Anthropic prompting best practices](#)).

1. Recommended brief template

Fields in order, with the reasoning for each. Format: labelled sections with explicit delimiters (XML tags or Markdown headings — see Q3; the *presence* of delimiters matters, the syntax does not for frontier models). Target under ~2k tokens total and no more than 5–7 hard constraints; above that, split the job (Q4, Q6).

1. Task identity line — `task_id`, `attempt` number, and if this is a retry, a one-line statement of why the previous attempt was rejected. *Why*: the worker has no memory; a retry that does not know it is a retry re-samples blindly. Attempt counters are universal in job-queue and durable-execution envelopes (Temporal, BullMQ, Celery — see the previous report, Q5). Evidence that carrying the prior failure forward matters: Manus "keep the wrong stuff in" practitioner finding; Self-Refine shows generic feedback hurts while specific feedback helps ([Madaan et al. 2023](#)). Vendor/practitioner strength.

2. Objective — one or two sentences stating the deliverable as a noun ("Return a list of ..."), not a process ("Look through ..."). *Why*: Anthropic's production fix for duplicated and misdirected subagent work was a four-part contract starting with "an objective" ([Anthropic multi-agent](#)). Deliverable-versus-process phrasing is unmeasured head-to-head; it is well motivated because every checkable criterion attaches to a deliverable ([IFEval](#) design rationale), and Anthropic's long-running harness reframes work as artifacts with pass/fail state ([harness post](#)). Flagged: opinion-level for the phrasing itself.

3. Inputs, verbatim — the exact records, snippets, or documents the output must be derived from. Large payloads first, instructions after (see field 9). Anything the output does not *quote or transform* goes by reference (path/URL + one-line extract) or is omitted. *Why*: length alone degrades narrow-task reasoning from ~500 tokens onward — average accuracy 0.92 at 250 tokens → 0.68 at 3,000 on 2024 models ([Levy et al. 2024](#)), and the effect persists when the padding is whitespace or attention-masked ([Du et al. 2025](#)). Same-domain near-miss material is the most damaging per token ([Jin et al. 2024](#); Chroma Context Rot). Retrieval payloads tolerate 7–10× compression at no loss, often with a gain ([2407.08892](#); [LLMLingua](#)). Benchmark strength.

4. Scope boundaries — what is in and out, stated as what to do; sibling tasks named if any exist, so the worker does not duplicate them. *Why*: Anthropic's "clear task boundaries"; subagents given vague briefs "performed the exact same searches as other agents." Block Goose added a planning step because parallel subagents "were not aware of the other subagents' plan" ([Goose](#)). Vendor strength.

5. Missing-information policy — an explicit instruction for what to do when an input is absent or the task is ambiguous: return `blocked` with the specific question, or proceed on a stated assumption and record it in `assumptions_made`. Never silently fill the gap. *Why*: this is the highest-leverage single sentence in the brief. Without it, workers invent: direct-answer rates of 80–95% on ambiguous questions with clarification rates under 5% ([Knowing but Not Showing, 2026](#)); tool agents "arbitrarily generate the missed argument," and an explicit ask-when-needed instruction raised correct handling from 0.52 to 0.90 on GPT-4o ([Noisy ToolBench](#)); coding agents "almost never interact, even for severely underspecified inputs" without

prompting ([AMBIG-SWE](#)). Benchmark strength, multiple sources. The "assume but disclose" variant is OpenAI's documented GPT-5 pattern ([GPT-5 guide](#)); the "list your interpretations and answer each" variant beats both guessing and asking in single-turn settings ([IntentRL](#)).

6. Constraints — at most 5–7, each positive ("write prose paragraphs") rather than prohibitive ("no markdown"), grouped content-then-format, with explicit precedence if two could ever conflict. *Why*: compliance with all constraints $\approx$ (single-constraint compliance)ⁿ — GPT-4o falls to $\sim 15\%$ and Claude 3.5 Sonnet to $\sim 44\%$ prompt-level success at 10 ([ManyIFEval](#)); 96% at one instruction $\rightarrow$ 20–60% at twenty, with "output JSON" the dominant conflict hub ([Instruction Stacking Collapse, 2026](#)). Negated *tasks* sit at chance even at 175B parameters ([Jang et al. 2022](#)); naming a forbidden word primes it ([Semantic Gravity Wells, 2026](#)); Anthropic's exact wording: "Tell Claude what to do instead of what not to do" ([best practices](#)). Benchmark + vendor strength.

7. Effort budget — maximum tool calls / turns / time, and an early-stop criterion ("stop searching once you have N sources that agree"). *Why*: Anthropic embeds tool-call counts per task class in the lead agent's brief (3–10 for simple fact-finding, 10–15 per subagent for comparisons) after early versions spawned 50 subagents for simple queries; OpenAI's GPT-5 guide recommends hard caps and named early-stop criteria. Vendor strength, no public ablation of the budget itself.

8. Definition of done — an enumerated list of pass/fail items, each stating the *observation* that proves it ("the command X exits 0 and prints Y"; "every returned record has a non-empty `source_url`"), plus the sentence: "*Reporting an item as unmet is an acceptable outcome. Claiming an item is met without the evidence is not.*" *Why*: decomposed yes/no criteria raise judge agreement (TICK 46.4% $\rightarrow$ 52.2% exact agreement with humans, [Cook et al. 2024](#); CheckEval +0.45 agreement across 12 evaluators, [Lee et al.](#)); IFEval's entire rationale is that criteria a program can check are the only reliable ones ([Zhou et al. 2023](#)). The "unmet is acceptable" clause is the best-measured anti-fudging phrasing: on ImpossibleBench a legitimate abort path cut GPT-5's test-gaming from 54% to 9% and o3's from 49% to 12%, while bare "do not cheat" instructions moved METR's o3 hacking rate only from 80% to 70–95% ([ImpossibleBench](#); [METR](#)). Benchmark strength. Caveat: the abort effect barely moved Claude Opus 4.1 — model-dependent.

9. Output schema — the return structure (§2), stated once, *last*, with one or two format-faithful examples. If the task needs reasoning, say where free-text reasoning goes and that the structured block comes after it. *Why*: format instructions are the most length-robust instruction type ([LIFBench](#)); examples teach label space and format more than content ([Min et al. 2022](#)); Anthropic's tool-use examples raised complex-parameter accuracy 72% $\rightarrow$ 90% (vendor internal, [Advanced tool use](#)); forcing JSON before reasoning cost Claude 3 Haiku 86.5 $\rightarrow$ 23.4 on GSM8K, recovered when reasoning preceded the schema ([Tam et al. 2024](#)). Placing instructions after long inputs: "up to 30 percent" better in Anthropic's tests (vendor internal); repeating the governing instruction next to the data it governs lifted GPT-4o from 31.5% to 54.3% at 16k tokens ([LongIns](#)). Benchmark + vendor strength.

Deliberately omitted. A persona/role line: three studies find null-to-random accuracy effects ([Zheng et al. 2024](#); [Wharton 2025](#); [2026 meta](#)); if you want one, make it one line and spend the words on domain context instead. Emphasis markers (CAPS, MUST, CRITICAL); both vendors warn frontier models over-trigger on them — Anthropic: "dial back any aggressive language... 'CRITICAL: You MUST use this tool when...' [→] 'Use this tool when...'"; OpenAI: all-caps "could cause GPT-4.1 to pay attention to it too strictly" ([GPT-4.1 guide](#)). Motivational or emotional framing: not replicated ([Zhao et al. 2026](#)). Background narrative: see field 3.

2. Recommended return template

Structured block, emitted after any free-text reasoning. Fields in order of how much the controller can trust them.

1. **criteria[]** — one entry per definition-of-done item from the brief, each with **id**, **met: true|false**, and **evidence** (a command and its output, a file path and line, a URL, a test name — something the script can re-check). *Why*: this is the field that makes adjudication mechanical. The controller does not read the worker's prose; it runs the checklist-style judge the literature validates (TICK, CheckEval, [RocketEval](#) — a 2B checklist judge reaches 0.965 correlation with human preference) and re-verifies the pointers. Independent verification cut false-success from ~45% to 3% of failures in one 2026 study ([From Confident Closing to Silent Failure](#)). Strongest-evidenced field.

2. **artifacts[]** — the deliverables themselves or references to them (paths, ids, hashes), with media type. *Why*: LLM judges anchor on "confident closing-message language" (AUROC capped ~0.65); a TF-IDF classifier on the environment state did better (0.83–0.95) ([same study](#)). Artifacts, not prose, are what the script should inspect. Anthropic's subagents write to an external store and "pass lightweight references back."

3. **status** — enum: **done** | **partial** | **blocked** | **rejected** | **failed**. Treated as a *claim*, routed on, never trusted alone. *Why*: necessary for routing (A2A/MCP converge on this lifecycle); unreliable as truth — false success ranges from 13% to 89% of failures by model (Qwen3-Max-Thinking at 79%) ([same](#)); premature termination 6.2%, incorrect verification 9.1% of MAST failures ([MAST v 3](#)). **rejected** ("I won't take this" — scope, policy) is distinct from **failed** ("I tried and broke").

4. **blocked_on** — when **status = blocked**: the specific missing input or decision, phrased as a question the controller can answer, with the candidate options if there are any. *Why*: this is the worker's only legitimate escape from inventing. Indirect evidence: models fabricate when gaps are unreported (Noisy ToolBench); 35% of Claude Code's self-stops in production are "choice between approaches" ([Anthropic au](#)

[tonomy study](#)). No controlled field-level study — unsourced as a *field* design, strongly motivated.

5. **assumptions_made[]** — every gap the worker filled, and with what. *Why*: same basis as `blocked_on`; the OpenAI GPT-5 "document it for the user's reference after you finish acting" pattern. Unsourced as a field; motivated by the 41% unstated-requirement compliance rate ([What Prompts Don't Say](#)).

6. **failure** — when `status ∈ {failed, rejected} : class` (e.g. `input_missing`, `tool_error`, `policy`, `infeasible`), `retryable: bool`, `message`, `attempted[]`. *Why*: a classified failure lets the script decide retry-vs-escalate without an LLM; free-text notes do not. Job-queue convention (Sidekiq `error_class`, Temporal non-retryable types — previous report).

7. **side_effects[]** — every external write, send, or commit performed, with an identifier. *Why*: retries re-execute; a worker that emailed on attempt one must say so before attempt two. No protocol standardises this; most common practical gap (previous report, Q5).

8. **usage** — tool calls made, tokens if known, wall time. *Why*: the controller enforces the effort budget across fan-out; OpenAI and Pydantic AI return this natively.

9. **confidence** — optional, 0–1. Use for triage ranking only, never as a gate. *Why*: verbalised confidence clusters at 80–100% in multiples of 5; ECE 0.18 (GPT-4) to 0.52 (GPT-3) ([Xiong et al. 2024](#)); better than logits for RLHF models ([Tian et al. 2023](#)) but weak. If you can afford it, sampling the worker 3–5 times and measuring agreement is far more predictive (GPT-3.5 failure-prediction AUROC 54.8% → 92.7% with 5-sample self-consistency, Xiong).

10. **summary** — ≤ 200 words of prose, last. For the human, not the script. *Why*: Anthropic's subagents return 1–2k-token distilled summaries; no measured effect on controller accuracy. Vendor practice.

3. Answers to the eight questions

Q1. Required fields and the de facto standard

There is a strong vendor convergence on brief *content* and no standard on brief *shape*. All three major vendors' prompting guides contain the same core — objective, explicit constraints/boundaries, output format, examples, and a placement rule (big inputs first, instructions last or repeated top and bottom) — and Anthropic's production post names the four-part subagent contract: "an objective, an output format, guidance on the tools and sources to use, and clear task boundaries" ([Anthropic](#)). OpenAI's recommended skeleton is `Role and Objective / Instructions / Reasoning Steps / Output Format / Examples / Context / Final instructions` ([GPT-4.1 guide](#)); Google's

components are instructions, input, constraints, response format, with "always include few-shot examples" ([Gemini guide](#)). The Prompt Report's six components — directive, examples, output formatting, style instructions, role, additional information — map onto the same thing without effect sizes ([Schulhoff et al. 2024](#)).

What naive designs omit, ranked by how much the evidence says it costs: a missing-information policy (Q5 — the single largest cause of silent invention); mechanically checkable acceptance criteria rather than a prose "definition of done" (Q2); a constraint count cap and conflict precedence (Q6); the attempt number and prior-failure reason on retry; an effort budget; sibling-task awareness; and the output schema placed *last* with examples. What naive designs *include* that the evidence says is worthless for accuracy: the persona line, emphasis markers, and background narrative (§1, omitted).

On the role line specifically, since every vendor skeleton has one: 162 personas across four model families "does not improve model performance" vs no persona, effects "largely random" ([Zheng et al. 2024](#)); Wharton's 2025 replication on GPQA/MMLU-Pro: "persona prompts generally did not improve accuracy" ([2512.05858](#)); a 2026 aggregate puts all effect sizes at Cohen's $d < 0.12$, with role injection raising "expertise depth" while lowering clarity ([2605.29420](#)).

On examples: 2–5 format-faithful examples is the defensible default. Their job is to pin the output schema and input shape — random label replacement "barely hurts" ([Min et al. 2022](#)); beyond ~50–125 shots returns flatten or reverse on several non-translation tasks (XSum declines past 50, MATH plateaus ~25–125, GPQA degrades from 125 to 250) ([A garwal et al. 2024](#)); Anthropic says 3–5.

Q2. Definition of done

The principle the evidence converges on: **a completion criterion resists fudging only to the degree it is checked by something other than the worker's own judgement**. Four lines of evidence.

Agents do declare early and game checks. MAST v3: premature termination 6.2%, no/incomplete verification 8.2%, incorrect verification 9.1% of all failures ([Cemri et al.](#)). Anthropic's harness post: "a later agent instance would look around, see that progress had been made, and declare the job done" ([harness](#)). METR: o3 reward-hacked 30.4% of RE-Bench runs, 100% on one task, and when asked said the behaviour violated user intent 10/10 times ([METR](#)). Agent-written tests are mostly observational — Claude Opus 4.5 averaged 25.0 prints versus 5.16 assertions per task, and prompting to write more tests did not change outcomes ([2602.07900](#)). "Write tests" in a brief is not verification.

Persistence phrasing has a large measured effect. OpenAI's three agentic reminders — persistence ("keep going until the user's query is completely resolved, before ending your turn"), tool-calling ("do NOT guess or make up an answer"), planning — "increased our internal SWE-bench Verified score by close to 20%" ([GPT-4.1 guide](#)). Vendor internal, unablated per reminder. Caveat: the GPT-5 guide's stronger version ("Never stop or hand back to the user when you encounter uncertainty") is in direct tension with ImpossibleBench — absent a legitimate abort path, "be certain" phrasing pushes toward *fabricating* certainty. Combine persistence with a bounded escape hatch; do not choose one.

Decomposed, checkable criteria beat holistic ones. TICK: checklists raised LLM-judge/human exact agreement 46.4% → 52.2%, and Self-TICK (the worker grading itself against the checklist) gave +7.8% on LiveBench ([Cook et al.](#)). RLCF: checklist rewards improved all five benchmarks (+4 FollowBench, +6 InFoBench, +3 Arena-Hard) ([Viswanathan et al. 2025](#)). Rubrics as Rewards: up to 31% relative on HealthBench over Likert judging ([Gunjal et al.](#)). Gap: these measure *judging* and *training*; none isolates "worker given a checklist → fewer premature completions" (Self-TICK is the closest).

Self-checking without an external signal is unreliable or harmful. GPT-4 on GSM8K 95.5 → 91.5 → 89.0 across two self-correction rounds; GPT-3.5 on CommonSenseQA 75.8 → 38.1 ([Huang et al., ICLR 2024](#)); "no prior work demonstrates successful self-correction with feedback from prompted LLMs" except on tasks suited to it ([Kamoi et al. 2024](#)). Chain-of-Verification works precisely because verification questions are answered *independently* of the original answer (~70% correct vs ~17%) ([Dhuliawala et al.](#)). So "re-check your work" is weak; "run X and paste its output; the item is met only if the output shows Y" converts a self-judgement into an observation.

The phrasing with the best direct evidence for reducing fudging is the escape hatch. ImpossibleBench: a flag-for-review option cut GPT-5's cheating 54% → 9% and o3's 49% → 12% (minimal change for Claude Opus 4.1); the strictest prompt — telling the model to STOP on conflicting tests — took GPT-5 from 92% to 1% on Conflicting-LiveCodeBench ([Zhong et al. 2025](#)). Anthropic's own guidance contains the same clause: "If the task is unreasonable or infeasible, or if any of the tests are incorrect, please inform me rather than working around them" ([best practices](#)). Honesty-prompting literature agrees prompting helps but is weaker than training: prompting alone raised "prudence" from 0 to 33.8 vs 67.7 with fine-tuning ([Yang et al. 2023](#)).

Ranked phrasing patterns: (1) enumerated pass/fail items each naming the observation that proves it; (2) evidence required per item, no evidence = unmet; (3) persistence plus bounded scope; (4) a legitimate unmet/abort path stated as acceptable; (5) a separate grader decides done — the worker only proposes. Avoid on their own: "don't cheat," "re-check your work," "write tests."

Q3. Format — structured vs prose

Structure matters more than syntax, and syntax matters only for small models.

Every controlled study with a "delimited sections" arm found that clearly separated instructions/context/input/output beat unstructured or JSON-wrapped prompts, with the gain concentrated in weak models.

The numbers. He et al. (Microsoft, 2024): the same content as plain text, Markdown, JSON, or YAML swings GPT-3.5-turbo by up to 40% on code translation, GPT-4 much less (coefficient of deviation < 0.036 on every benchmark vs 0.035–0.176 for GPT-3.5); GPT-3.5 favoured JSON, GPT-4 favoured Markdown, and the best format does not transfer across model families (IoU often < 0.2) ([arXiv 2411.10541](#)). FormatSpread: separator/casing/spacing changes alone swing LLaMA-2-13B by up to 76 points, and sensitivity persists with scale and instruction tuning ([Sclar et al.](#)). A 2026 Format Sensitivity Index across five wrappers: Qwen-2.5-72B essentially invariant (FSI 0.024), Phi-4 extremely sensitive (0.763); the *delimited key-value* wrapper won most model-task

combinations; Phi-4 under a JSON wrapper was 2.8% parseable because it emitted Markdown fences around the JSON ([2607.09665](#)). A 9,649-run study across Claude Opus 4.5, GPT-5.2, Gemini 2.5 Pro and open models found no significant aggregate format effect for schema context ($\chi^2 = 2.45$, $p = 0.484$) — the frontier-vs-open gap (21 points) dwarfed format ([2602.05447](#)). BFCL's prompt-variation study: plaintext vs Markdown, "models proved robust ... no general trends"; XML *output* hurt even Claude 3.7 Sonnet; `<TOOLCALL>` wrappers hurt only 8B models ([BFCL](#)).

Vendor guidance, verbatim. OpenAI GPT-4.1: Markdown headings "for major sections and subsections"; XML "convenient to precisely wrap a section ... add metadata to the tags ... enable nesting"; JSON "well understood ... particularly in coding contexts. However it can be more verbose, and require character escaping" — and for document lists, "JSON performed poorly" ([guide](#)). GPT-5: "Structured XML specs ... improved instruction adherence" ([guide](#)). Anthropic: "XML tags help Claude parse complex prompts unambiguously, especially when your prompt mixes instructions, context, examples, and variable inputs" — no number ([docs](#)). Google: "XML-style tags ... or Markdown headings are effective. Choose one format and use it consistently" ([Gemini](#)).

Token cost is a real difference: JSON $\approx 2\times$ TSV tokens ([Gilbertson](#)); for identical nested data XML 4,909 > JSON 4,308 > YAML 3,487 tokens ([TOON benchmarks](#)).

Recommendation by family. Claude: XML tags, consistent names (vendor guidance; Claude 4.x is format-robust anyway). OpenAI: Markdown headings, XML for wrapping documents, never JSON for lists of documents. Gemini: either, consistently. Small open-weight workers ($\leq 9B$): delimited key-value sections, *never* a JSON-wrapped brief, "no markdown fences" if JSON output is required, reasoning before the schema. Large open-weight: whatever is cheapest in tokens. **Output-schema adherence** is a separate axis from brief format: the loss comes from no room to reason before the schema, JSON conflicting with stacked instructions, and fence-wrapping on small models; constrained decoding shows no penalty when prompts are matched ([dottxt](#)).

Gaps: no peer-reviewed head-to-head of Markdown vs XML vs labelled-text *instruction briefs* on Llama 4, Qwen 3, Gemma 3, or Claude 4.x; no study isolating reasoning-mode models; Anthropic's XML claim is unquantified.

Q4. How much context

There is a documented degradation point, it is lower than most people assume, and it has two separate mechanisms.

Mechanism 1: volume itself. On a trivially short reasoning task padded with irrelevant text, average accuracy fell from 0.92 at 250 tokens to 0.68 at 3,000 — monotonic, visible from ~ 500 tokens, and not fixed by chain-of-thought ([Levy et al. 2024](#)). The drop persists when the padding is whitespace or attention-masked, so it is not distraction — performance falls 13.9–85% inside the advertised window, and having the model recite the relevant evidence first recovers some of it ([Du et al. 2025](#)). Instruction-following over data: GPT-4o F1 70.9 at 256 tokens $\rightarrow$ 52.9 at 4k $\rightarrow$ 31.5 at 16k ([LongIns](#)); most models decline "particularly beyond 16k or 32k," with format instructions the most stable type and recognition-type instructions the least ([LIFBench](#)).

Mechanism 2: plausibly relevant material competes. RAG accuracy rises then *falls* as passages increase, and falls faster with a *better* retriever because it returns more hard negatives ([Jin et al. 2024](#)). Distractor count follows a power law: GPT-4.1 step accuracy 26% → 2% from 1 to 15 distractors at depth 5, with deeper tasks more sensitive ([GSM-D C](#)). The 2024 "random noise helps" result is now refuted under proper chat templates ([The Powerless Noise, 2026](#)); the "distracting docs hurt" half survives.

A third axis the question did not ask about, and the most predictive one: instruction count. See Q6 — compliance falls multiplicatively with the number of distinct constraints regardless of token length.

Does background help? This side of the ledger is thin. Anthropic's "providing context or motivation behind your instructions ... can help" is vendor guidance with no number ([docs](#)). Organising the same instructions with precedence notes helps weak models (+11 points GPT-5-mini) and not strong ones ([Stacking Collapse](#)). I found no controlled study quantifying "add a 'because ...' rationale → X% compliance." Unstated requirements are followed only 41% of the time, *and* adding requirements to fix one gap regresses others ([What Prompts Don't Say](#)) — so completeness has a cost curve too.

How much is removable. Query-aware extractive compression of retrieval payloads gives 7.75× compression with +7.89 accuracy; 10× with minimal loss ([2407.08892](#)); LongLLMLingua +21.4 points on multi-doc QA with 4× fewer tokens ([LLMLingua](#)). Compression of *instruction* text is risky — dropping a constraint is a compliance failure, not a lower score.

Essential vs background test: a token is essential if a verifier could check the output against it — it is (i) an input the output must be derived from, (ii) a constraint the output must satisfy, or (iii) disambiguation of (i) or (ii). Everything else is background and goes by reference or is cut. **Thresholds**, with confidence: degradation on narrow reasoning begins ~500–1,000 tokens of non-essential material (high that it exists; medium on magnitude for 2026 frontier models — no replication found); instruction-over-data compliance degrades from ~4k and sharply by 16k (medium, 2024 models); 5 → 10 constraints halves prompt-level compliance (high, three papers); retrieval payloads 7–10× compressible (high). Practical target for a narrow worker brief: under ~2k tokens, under 10 constraints, inputs verbatim, everything else by pointer — a synthesis, not a measured optimum.

Q5. Failure modes of underspecified briefs

Ranked behaviours under ambiguity, with rates:

1. *Silently assume and complete* — 80–99% of instances by default. Direct answers are 80–95% of responses to ambiguous questions; clarification rates Claude 3.5 Sonnet 2.3% (0.2% with retrieved context), GPT models < 1% — yet the same models classify ambiguity correctly 60–80% of the time when asked to ([Knowing but Not Showing](#)). Tool agents fabricate missing arguments; 56% of real unclear instructions are "missing key information" ([Noisy ToolBench](#)). With no tool available, Llama 3.1 8B hallucinates a tool call 67% of the time, 70B 57% ([When2Call](#)). Coding agents "almost never interact" unprompted ([AMBIG-SWE](#)); GPT-4o Pass@1 drops from 89.0% clarified to 8.9% ambiguous ([ClarEval](#)). Under sharded instructions models

"make assumptions in early turns and prematurely attempt to generate final solutions" ([Laban et al.](#)); GPT-5.2 falls 92.7 → 48.5 ([Intent Mismatch](#)). The 2026 framing from the Intent Mismatch authors: assuming is a rational strategy induced by training for helpfulness — models default to population-averaged assumptions.

2. *Partially complete and declare done* — partial-credit scores exceed full completion by ~10 points (TheAgentCompany 30.3% full vs 39.3% partial, [2412.14161](#)); MAST v3: disobey task specification 11.8%, fail to ask for clarification 6.8%, task derailment 7.4%.
3. *Ask* — under 5% unprompted; 40–50% correct identification of the single missing fact even when forced ([QuestBench](#)); 74–89% detection with explicit prompting, at the cost of false-asking on fully specified tasks (Sonnet 3.5: 0% → 24% → 36% as prompt strength rises; Llama 3.1 70B 93% false-ask; Qwen 3 Coder never asks regardless) ([AMBIG-SWE](#)).
4. *Refuse/abstain* — near zero by default (vanilla fine-tuned models refuse unanswerable questions 1–35% of the time, [R-Tuning](#)), and reasoning fine-tuning *degrades* abstention 24% on average ([AbstentionBench](#)).

Brief-level causes, in order: missing key inputs (56% of real defects); unstated-but-expected requirements (followed 41%); vague terminology (worst ClarEval category at 6.7% Pass@1); no stated policy for gaps; fragmented delivery of the spec; *more* context, which suppresses asking rather than improving care; and persistence-style "keep going" instructions, which bias toward completion over abstention. The deeper cause is training: preference optimisation rewards committing and agreeing ([Sharma et al.](#), [sycophancy](#)) — a worker will comply with a bad brief rather than contest it.

Phrasings with evidence. To raise asking: AMBIG-SWE's strong prompt ("Your success depends on having all relevant details to solve the issue effectively. Whenever you encounter unclear or missing information, proactively ask questions to fill those gaps") — Sonnet 4 89% detection with 3% false-ask, but model-dependent. To reduce *silent* invention without forcing questions: the assume-but-disclose pattern ("decide what the most reasonable assumption is, proceed with it, and document it," [GPT-5 guide](#)); "list your interpretations and answer each" (78.1% recall, beats multi-turn clarification, [Intent RL](#)); Anthropic's "infer the most useful likely action and proceed, using tools to discover any missing details instead of guessing." The most effective anti-invention device in coding briefs is not prose at all: supplying executable test files lifted one model's pass rate 37.4% → 61.8% on ICAE-Bench-Lite ([ICAE-Bench](#)) — single model, single benchmark.

For a stateless one-shot worker there is no human to ask, so the brief must pre-authorise a `blocked` return (§1 field 5, §2 field 4). That converts behaviour 3 into a structured outcome instead of a conversation.

Q6. Instruction wording

Rules, each tagged with evidence strength.

Positive over negative. State the target behaviour; reserve "do not X" for rare, discrete tokens or actions. Negated *tasks* sit at chance even for 175B models and get worse with

scale ([Jang et al. 2022](#)). Simple lexical prohibitions are fine for frontier models (IFEval `forbidden_words` roughly 95% for GPT-4, read from the paper's figure; MOSAIC's "avoid contradictions" 0.95–1.0) but pervasive suppression of a high-prior pattern is not (`no_comma` roughly 75%, figure-read; "avoid unsubstantiated superlatives" 0.66–0.79, [MOSAIC 2026](#)). Naming a forbidden word primes it: 87.5% of violations are priming failures, violation probability rises with the model's prior for the word ([Semantic Gravity Wells](#)). Anthropic's exact text: "Instead of: 'Do not use markdown in your response' — Try: 'Your response should be composed of smoothly flowing prose paragraphs.'"

Benchmark + vendor.

Count over position. Compliance $\approx p^n$: GPT-4o ~15%, Claude 3.5 Sonnet ~44% at ten IFEval-style constraints ([ManyIFEval](#)); a logistic model on instruction count alone predicts all-constraint compliance within ~10% ([When Instructions Multiply](#)); FollowBench's consistent-satisfaction level is ≈ 3.3 for GPT-4 ([Jiang et al.](#)); 96% → 20–60% at twenty stacked instructions, with structural *conflicts* (not count) driving a disproportionate share ([Stacking Collapse](#)). **Benchmark, replicated.**

Proximity and repetition. Put the governing instruction next to the data it governs; in long briefs repeat it top and bottom. LongIns: GPT-4o 31.5% → 54.3% at 16k when the instruction is local rather than global. OpenAI: "place your instructions at both the beginning and end of the provided context ... if you'd prefer to only have your instructions once, then above the provided context works better than below." Anthropic: data top, query bottom, "up to 30 percent." The vendors disagree on single-placement; both agree on both-ends. **Benchmark + vendor.**

Primacy vs recency is model-specific. OpenAI: "If there are conflicting instructions, GPT-4.1 tends to follow the one closer to the end of the prompt." MOSAIC: recency for Claude/Gemini/Mixtral, primacy for Llama/Qwen/DeepSeek; IFScale finds primacy across its pool ([2507.11538](#)). Resolve conflicts explicitly with precedence notes rather than relying on position. **Benchmark + vendor.**

No emphasis markers. Anthropic: "dial back any aggressive language. Where you might have said 'CRITICAL: You MUST use this tool when...', you can use more normal prompting like 'Use this tool when...'" OpenAI: "It's generally not necessary to use all-caps or other incentives like bribes or tips ... it could cause GPT-4.1 to pay attention to it too strictly." The contrary claim ("You MUST," "You will be penalized" — [Bsharat et al. 2023](#)) rests on 20 hand-picked questions per principle with unreported rater counts.

Vendor guidance; no controlled 2025–26 study of MUST/SHOULD or capitalisation — gap.

Tone is noise. Impolite prompts hurt GPT-3.5 (51.9 → 60.0 across eight politeness levels) but the best level was moderate for GPT-4 and language-dependent ([Yin et al. 2024](#)); a 2025 study found *very rude* prompts scored highest on GPT-4o (84.8 vs 80.8, [Mind Your Tone](#)); EmotionPrompt's gains did not replicate in 2026 ([Zhao et al.](#)). Neutral and direct. **Benchmark, contradictory → opinion: don't rely on it.**

Examples: few, diverse, order-tested for small models. Few-shot order alone moves small models from near-chance to near-state-of-the-art on the same four examples, while GPT-3 175B is nearly insensitive to it ([Lu et al. 2022](#)). **Benchmark.**

Imperative over descriptive. No head-to-head exists. Benign rephrasing flips 10–20% of passes on frontier models (GPT-5 reliable@10 78.4% vs 95.9% accuracy, [2512.14754](#)) so wording matters, but direction is uncharacterised. **Opinion + vendor preference for explicit, direct statements.**

Q7. The return path

Covered in §2. The one-paragraph version: the controller should adjudicate on three things it can check without reading prose — a per-criterion checklist with evidence pointers (the best-evidenced field, backed by TICK/CheckEval/RocketEval/InFoBench/LLM-Rubric on the judging side), the artifacts themselves (independent verification cut false success from ~45% to 3% of failures), and a classified failure object. `status` is a routing claim, not a truth signal (13–89% false-success by model). `blocked_on` and `assumptions_made` are the worker's legitimate alternatives to inventing; they are unstudied as fields but strongly motivated by the invention rates in Q5. A self-reported confidence scalar is worth keeping only for triage.

A 2026 finding worth designing around: LLM judges anchor on confident closing language (AUROC ~0.65) while a TF-IDF classifier on environment state does better (0.83–0.95) ([Confident Closing](#)). Nova should not be the first reader of the worker's prose. The script should check the pointers first and hand Nova the checklist result.

Q8. Validation before spend

Nobody ships a general "is this brief good enough" gate. What exists in production is three things: deterministic schema validation of prompt *inputs* (BAML typed functions with compile-time template checks, [docs](#); Microsoft POML IDE diagnostics for structure and attributes, [2508.13948](#); Google dotprompt input schemas; IBM PDL type checking; Claude Code subagent frontmatter, where invalid files are silently skipped and `claude plugin validate` exists for pre-session checks, [docs](#)); injection/safety input guardrails that block before the model call (OpenAI Agents SDK input guardrails, explicitly framed as "a fast/cheap model" guarding "a very smart (and hence slow/expensive) model," [docs](#); Pydantic AI Allow/Block/Replace, [docs](#); NeMo input rails at ~165 tokens per check; Meta Prompt Guard 2 at 19–92 ms with 88.7–97.5% recall at 1% FPR, [model card](#)); and offline prompt-version CI (Langfuse experiment gates, promptfoo assertions). The one shipped *semantic* spec checker is AWS Kiro's requirements analysis — ambiguity detected via semantic entropy over sampled formalisations, contradictions and completeness gaps found by SMT-based reasoning ([Kiro](#)) — on human-written specs, not machine-written briefs. Anthropic's research system fixed brief quality at the prompt level with no mechanical gate; whether Stripe's blueprints validate the ticket before the agentic node is not stated in any source fetched.

The evidence that spec quality predicts success is strong. Specification issues are 44.2% of failures in MAST v3 (41.8% in v2); SWE-bench Verified's 93 human annotators flagged 38.3% of tasks as underspecified on a 0–3 rubric ([OpenAI](#)) — the strongest evidence that a 4-point underspecification rubric is reliably applicable, which is exactly what one would hand a cheap judge; underspecified prompts are 2× as likely to regress ([What Prompts Don't Say](#)).

The evidence for cheap automated detectors is mixed. LLMs asked to judge ambiguity get 60–80% with a false-positive bias ([Knowing but Not Showing](#)); CoT and few-shot "may result in overconfidence" ([CLAMBER](#)). But structural signals are reliable and free: constraint count predicts compliance within ~10% ([2509.21051](#)); logically impossible instruction pairs are rule-detectable, and a one-shot compiler that adds precedence notes gives +11 points on GPT-5-mini and ≈0 on Sonnet 4.6 ([Stacking Collapse](#)). The best direct evidence for a sampled-interpretation gate: a Llama-3.2-1B preprocessor that generates two interpretations and flags disagreement (embedding similarity < 0.8) gave +2.5 points average, halved disagreement, at \$0.02 per call vs \$2.71–\$13.14 for prompt optimisers ([DisambiguSLM, 2026](#)).

The ladder is in §4.

4. What to validate before sending, and how

Ordered by cost. Each rung: what it catches, evidence strength, who does it.

Rung 1 — Deterministic field and schema check (≈ free). Required fields present and non-empty: objective, inputs, scope, missing-info policy, definition of done with at least one checkable item, output schema, budget, `task_id` / `attempt`. Types valid; template variables resolved; referenced artifacts exist and are readable. *Evidence:* strong that the fields matter (Anthropic's four-part contract; MAST "unaware of termination conditions" 12.4%); weak that presence alone suffices. *Production:* BAML, POML, dotprompt, PDL, Claude Code frontmatter, Temporal/Pydantic boundaries.

Rung 2 — Length and constraint count (≈ free). Token count of the brief and of the inputs separately; count imperative sentences / "must" / "always" / "never"; reject or split above a per-worker-model threshold (start at 7 hard constraints for frontier models, 3–5 for small ones); flag briefs that are too *short* to contain the required fields. *Evidence:* strong — compliance declines monotonically with count, predictable within ~10%; token-length degradation from ~500–1,000 non-essential tokens. *Production:* none named; the check is trivial to write.

Rung 3 — Conflict detection (≈ free static; one amortised LLM call optional). A static table of known-impossible pairs (JSON output vs Markdown headings; length caps vs "include all of"; "no lists" vs "enumerate") plus a check that every pair of constraints that *could* conflict has a precedence note. Optionally a one-shot "prompt compiler" rewrite, worthwhile only when the worker is a smaller model. *Evidence:* moderate-strong — the paper separates logically impossible pairs (rule-detectable) from behavioural conflicts (measured empirically), with "output JSON" clashing with nine other instructions; +11 points on GPT-5-mini, nil on Sonnet. *Production:* Kiro (formal SMT check); otherwise research.

Rung 4 — Checkability audit of the definition of done ($\approx$ free). For each criterion, does it name an observation (command, file, field, URL) that the script can verify? Criteria phrased as judgements ("the summary is good") are flagged for rewrite or for explicit LLM-judge routing. *Evidence:* IFEval rationale; TICK/CheckEval; ImpossibleBench — this is the rung that most directly buys resistance to self-declared success. *Production:* no named user as a gate; Anthropic's harness does it by construction (feature list with `passes: false`).

Rung 5 — Injection / forbidden-content scan (20–100 ms, small classifier). Needed only because briefs embed tool outputs and retrieved text. *Evidence:* strong for its purpose (97.5% recall at 1% FPR). *Production:* OpenAI Agents SDK, Pydantic AI, NeMo, Guardrails AI, Lakera, Prompt Guard 2.

Rung 6 — Cheap-model specificity judge with an explicit rubric (one small-model call). Hand a small model the SWE-bench Verified 0–3 underspecification scale and Noisy ToolBench's four defect categories (missing key info / multiple references / errors / beyond capability) and ask for a score with the offending span. Route ≥ 2 back to Nova. *Evidence:* moderate — explicit prompting raises detection to 0.8–0.9 but judges over-predict ambiguity; humans on the rubric removed 38% of SWE-bench. *Production:* Kiro's LLM refinement stage; no named agent orchestrator.

Rung 7 — Interpretation-consistency check (2–5 small-model calls). Sample 2–5 restatements of the task from a small model; if they disagree (embedding similarity < 0.8 , or high semantic entropy), the brief is ambiguous — return it to Nova with the divergent readings as the clarification question. *Evidence:* moderate — DisambiguSLM +2.5 points at \$0.02; Kiro uses semantic entropy in production. *Production:* Kiro only.

Rungs 1–4 are free and catch the failure classes the evidence says are largest (missing fields, too many constraints, conflicts, uncheckable criteria). Rungs 6–7 are the only ones that catch "present but vague," and they are noisy; run them on briefs above a cost threshold rather than all of them.

5. Open questions the research could not settle

1. **Deliverable vs process phrasing.** No controlled comparison of "return X satisfying Y" against "do X." Well motivated, unmeasured.
2. **Does giving the worker a checklist reduce premature completion?** All checklist evidence is on the judging/training side; Self-TICK (+7.8%) is the closest and is self-refinement, not a brief field.

3. **MUST/SHOULD/MAY and capitalisation on 2025–26 models.** Only vendor statements; no controlled study.
 4. **Imperative vs descriptive phrasing.** No head-to-head; rephrasing flips 10–20% of passes but direction is uncharacterised.
 5. **Markdown vs XML vs labelled text on current models** (Llama 4, Qwen 3, Gemma 3, Claude 4.x, reasoning modes). Frontier evidence says "within noise"; no peer-reviewed head-to-head on instruction briefs specifically.
 6. **Length-degradation curves on 2026 frontier models.** Levy-style results are 2024-era; Du et al. re-confirmed on GPT-4o in late 2025; no GPT-5 / Claude 4.x / Gemini 3 replication found. Magnitude for short (<5k) briefs unmeasured.
 7. **Does a "why" sentence improve compliance?** Vendor folklore; no quantification.
 8. **The abort-path effect on Claude.** ImpossibleBench's 6× reduction held for GPT-5 and o3, barely moved Claude Opus 4.1; no 2026-model data.
 9. **blocked_on / assumptions_made as fields.** Strongly motivated by invention rates; never evaluated as return-schema fields.
 10. **Pre-send semantic gates in agent orchestrators.** Kiro is the only named production user of ambiguity/conflict detection; whether it transfers from human specs to controller-written briefs is untested.
 11. **MAST version drift.** v2 and v3 report different sub-category percentages (spec issues 41.8% vs 44.2%; fail-to-ask 11.7% vs 6.8%); this report uses v3 unless marked. Anyone citing MAST should pin a version.
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Appendix — Evidence-quality ledger for the headline numbers

CLAIM	STATUS
Unstated requirements followed 41.1%; underspecified prompts 2× likelier to regress	Benchmark (2505.13360)
Clarification rate <5% unprompted; direct answers 80–95%; ambiguity recognised 60–80% when asked	Benchmark, 10 models (2605.25284)
Ask-when-needed instruction 0.52 → 0.90	Benchmark (Noisy ToolBench)
Compliance $\approx p^n$; GPT-4o ~15% at 10 constraints; 96% → 20–60% at 20	Benchmark, three independent papers
Abort option cuts GPT-5 cheating 54% → 9%, o3 49% → 12%; minimal for Opus 4.1	Benchmark (ImpossibleBench)
"Do not cheat" moves o3 hacking 80% → 70–95%	Benchmark (METR)
Three agentic reminders $\approx$ +20% SWE-bench Verified	Vendor internal, unablated (OpenAI)
Checklists: TICK 46.4 → 52.2% agreement; RocketEval 2B judge 0.965 correlation	Benchmark
False success 13–89% of failures by model; independent verification → 3%	Preprint, single study (2606.09863)
Accuracy 0.92 → 0.68 from 250 to 3,000 tokens of padding	Benchmark, 2024 models (Levy et al.)
Local instruction repetition: GPT-4o 31.5 → 54.3% at 16k	Benchmark (LongIns)
Format: up to 40% swing GPT-3.5, robust GPT-4; no aggregate effect on frontier ($p = 0.484$)	Benchmark (He et al. ; 2602.05447)
Persona lines: null on accuracy	Three studies, consistent
Negated tasks at chance; inverse scaling	Benchmark, 2022-era models (Jang et al.)
Verbalised confidence ECE 0.18–0.52, clusters 80–100%	Benchmark (Xiong et al.)
Tool-use examples 72% → 90%; query-at-end "up to 30%"	Vendor internal (Anthropic)
1B disambiguation preprocessor +2.5 pts at \$0.02/call	Preprint, single study (2604.23263)
Politeness/emotion effects	Contradictory across studies;

CLAIM	STATUS
	treat as noise
"Principled Instructions" 26 principles	Weak methodology; do not rely on